

Figure 1.27 Warp knitting consists of multiple yarns that loop together and interlock vertically

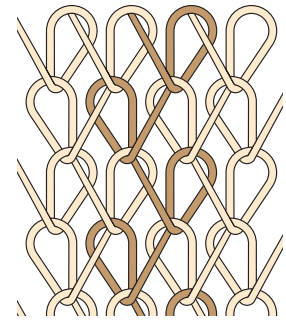


Figure 1.28 Weft knitting consists of one yarn with a series of loops that interlock horizontally

1.11.6 Properties

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Fibres have different properties and characteristics depending on the source of fibre and how the fabric has been constructed. This affects what it can be used for. Table 1.12 shows the performance properties of different fibres. The higher the star rating, the greater the property in the stated fibre.

Table 1.12 Performance properties of fibres

	Wool	Cotton	Polyester	Acrylic
Elasticity	****	*	****	***
Resilience	****	*	****	***
Durability	**	**	****	***

Now test yourself

TESTED

1 Name **one** natural fibre and **one** synthetic fibre. (2)
2 Explain how the type of weave in fabric construction affects the fabric. (2)
3 State **one** characteristic of knitted fabric. (1)
4 Explain why bonded fabrics are often used in disposable products. (2)
5 Describe **two** reasons why natural fibres are considered more environmentally friendly. (2)

1.12 Natural and manufactured timbers

- **Natural timber** comes from trees that are cut down into logs and then sawn into planks.
- **Manufactured timber** is made from recycled wood chips and fibres.

Natural timber: timber that comes from trees.

Manufactured timber: a human-made board made from recycled timber.

1.12.1 Natural timbers – hardwoods

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- **Hardwoods** come from deciduous trees that shed their leaves every year.
- Hardwood trees grow slowly and can take hundreds of years to grow fully.
- They have thick trunks with branches at the top.
- Hardwoods have a close **grain** and tend to be denser and heavier than softwoods.

Hardwood: timber that comes from deciduous trees.

Grain: the longitudinal arrangement of wood fibres.

Oak

- Oak is a heavy, hard and tough timber.
- It has an open grain and an attractive appearance.
- Oak is good for outdoor use as it is durable and can withstand the elements well compared to other woods.
- Common uses for oak include garden furniture, doors, floors and high-end furniture.

Mahogany

- Mahogany is a dark coloured close-grained timber.
- It is easy to cut and shape, is available in wide planks and polishes well.
- Common uses for mahogany include furniture, shop fittings, boat building and doors.

Beech

- Beech is a very tough, hard timber with a close and straight grain.
- It is hardwearing, easy to work and polishes well.
- It has a light brown colour and can be prone to warping.
- Common uses for beech include furniture, toys and wooden tools.

Balsa

- Balsa is a soft and lightweight timber with a coarse and open grain.
- It is very easy to shape, sand, glue and paint.
- Common uses for balsa include model making, packing cases, surfboards and fishing floats.

Typical mistake

Balsa is a hardwood, despite it being a very soft, lightweight and easy to shape.

1.12.2 Natural timbers – softwoods

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- **Softwoods** come from coniferous (evergreen) trees that have needles instead of leaves.
- Softwood trees keep their needles year-round and grow much faster than hardwood trees.
- They grow tall and straight, and have lots of branches all the way up the trunk.
- Softwood has more knots than hardwood.

Softwood: timber that comes from coniferous trees.

Pine

- Pine is the most widely available natural timber and is relatively cheap compared to other timbers.
- It has a straight grain and is easy to cut and shape.
- Pine has knots along the grain of the wood. They can weaken its strength but give an attractive appearance when finished.
- Common uses for pine include interior construction work, boxes and crates, flooring and paper (pulpwood).

Cedar

- Cedar is a lightweight, soft timber with a straight grain.
- It is knot free and very durable.
- Cedar is resistant to rot, weather and insect attack.
- Common uses for cedar include outside joinery, building cladding, bathroom and kitchen furniture, and wall panels.

Exam tip

Hardwoods are generally heavier, harder, tougher and more durable than softwoods.

1.12.3 Manufactured timbers

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- Manufactured boards can be made from hardwood and softwood.
- They are made by gluing and compressing wood fibres or layers together.
- They are made in large sheets that are easy to work with.
- Manufactured boards are generally cheaper than 'real' or natural wood.



Figure 1.29 Plywood is made from thin layers of timber placed with alternating grain direction

Plywood

- Plywood is made by gluing together thin layers of timber with alternating direction of grain.
- It is strong and tough due to its layered construction, with a high strength-to-weight ratio.
- Plywood is easy to cut and shape but can splinter easily.
- It is relatively stable under changes in temperature and moisture.
- Common uses for plywood include structural panelling in building construction and furniture making.

Medium-density fibreboard

- Medium-density fibreboard (MDF) is made by gluing and compressing wood fibres together into sheets.
- It has a smooth, even surface but chips and damages easily.
- It is easy to cut, shape, paint, glue and stain.
- MDF has poor moisture resistance and degrades quickly when wet.
- Common uses include cheap flat-pack furniture, interior panelling and interior doors.

1.12.4 Properties

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Hardness

- **Hardness** is the resistance of the wood to denting and wear.
- It varies depending on the direction of the wood grain.
- Hardness of wood is tested by dropping a ball bearing onto it from a set height. This is called the Janka hardness test.

Hardness: resistance to impact and abrasion.

Toughness

- Toughness is the amount that timber can be bent without breaking or splitting.
- Toughness is affected by the timber's internal cell structure.
- Most hardwoods have interlocking cell structures and are generally tougher than softwoods.

Durability

- **Durability** is the wood's ability to resist decay from exposure to the elements and attack from insects.
- Hardwoods such as teak are the most durable woods and can last for many years when exposed to the elements.
- Some types of bamboo have a durability of only a few months.

Durability: resistance to wear, long lasting.

Now test yourself

TESTED

1 Why is plywood a much stronger manufactured board than MDF? (2)
2 Name **one** advantage and **one** disadvantage of knots in pine. (4)
3 Explain **one** major difference between hardwood and softwood trees. (2)

1.13 Contexts which inform design outcomes

1.13.1 Discriminating between materials, components and manufacturing processes

REVISED

Properties may change depending on things such as temperature or the shape and direction of the material. Other properties will remain the same regardless of the environment or conditions.

Mechanical properties include:

- compressive strength – resistance to a pushing force
- tensile strength – resistance to a pulling force
- durability – resistance to wear, long lasting
- hardness – resistance to abrasion
- stiffness – resistance to bending and flexing
- elasticity – ability to stretch and return to shape
- impact resistance – resistance to knocks and impacts
- brittleness – hard but easily broken or shattered
- water resistance – resistance to water or moisture
- absorbency – ability to soak up moisture.

Manufacturing properties include:

- machinability – ability to be easily milled and drilled
- workability – ability to be placed and compacted
- ductility – ability to stretch, not break under stress or impact
- malleability – ability to be hammered, pressed or rolled without breaking.

Other properties include:

- acoustic – acoustic absorption
- chemical – corrosion resistance
- electrical – conductivity, resistance and so on
- thermal – conductivity, flash point and so on
- magnetic – attractive to magnets.

Typical mistake

Malleability and ductility are similar properties and often confused. Ductility relates to the ability of a material to stretch without damage. Malleability relates to the ability of a material to deform under compression.

Exam tip

By learning the properties of materials, you can understand why they have been used to make certain products. This will help you to answer questions in the exam about individual products and their fitness for purpose.